

Mobile Application Development
Assignment 6 (100 Points)

Basic Instructions:

1. In every file submitted you MUST place the following comments:
 - a. Assignment #.
 - b. File Name.
 - c. Full name of the student.
2. Each group is required to submit the assignment on Canvas.
3. **Submit Codes:**
 - a. Zip all the project folder to be submitted on canvas.
4. Submission details:
 - a. The file name is very important and should follow the following format:
Assignment#.zip
 - b. You should submit the assignment through Canvas: Submit the zip file.
5. **Failure to follow the above instructions will result in point deductions.**

In this assignment you will be building an application that uses ListViews to display books. The app requirements are as follows:

1. You are provided with a skeleton app that includes all the required UI and classes needed to complete this assignment.
2. All communication between fragments should be performed through the Main Activity.
3. When the Main Activity starts it should load the “Genres” Fragment as the initial fragment.

Part 1, Genres Fragment (30 Points):

This fragment displays the genres as shown in Figure 1(a). Please follow the steps:

1. Use a ListView to display the list of genres. The list of genres should be retrieved from the Data class.
2. Use a simple adapter to display the genre string values as shown in Figure 1(a).
3. Clicking a row item should communicate with the Main Activity to:
 - a. Send the selected genre and replace the current fragment with the Books fragment.
 - b. Push the current fragment on the back stack.

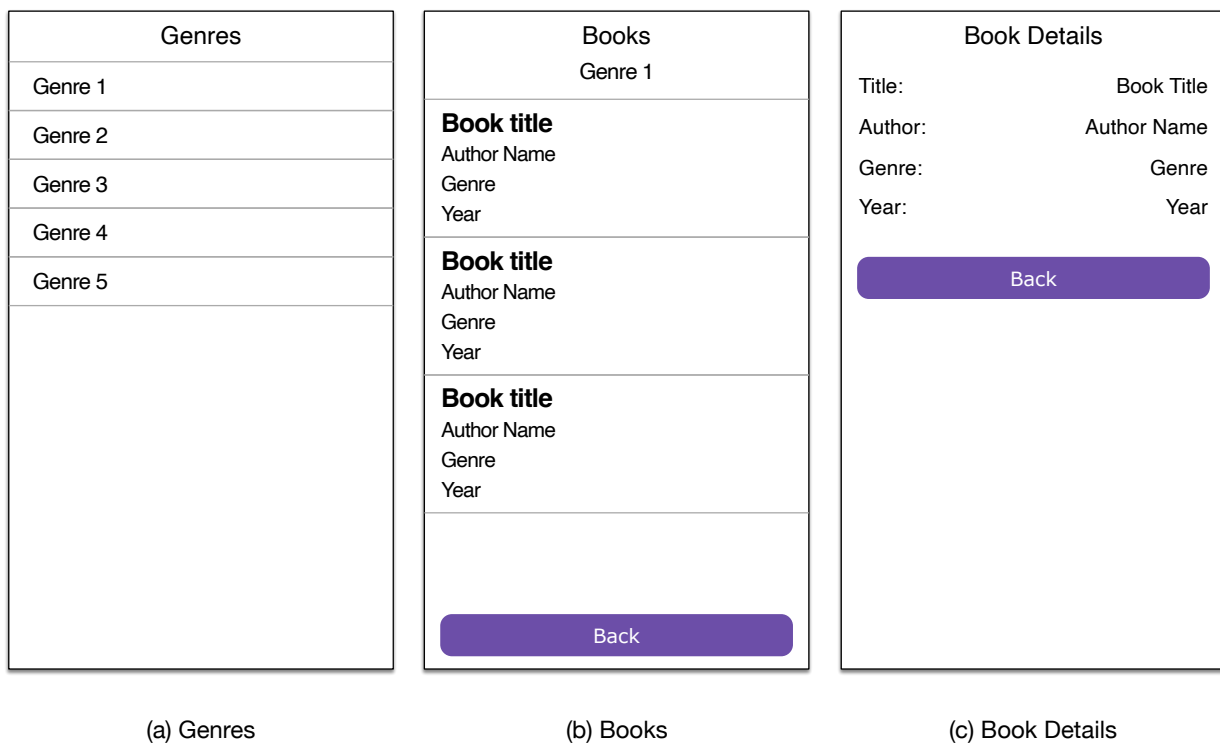


Figure 1, App Wireframe

Part 2, Books Fragment (60 Points):

This fragment displays the genres as shown in Figure 1(b). Please follow the steps:

1. This fragment receives the selected genre, which should be displayed as the title as shown in Figure 1(b). Get the list of books for the selected genre using the Data class.
2. Use a ListView to display the list of books for the selected genre. Each row item should show the book title, author, year and genre.
 - a. Create a custom adapter in the Books Fragment, which extends the ArrayAdapter.
3. Clicking "Back" should:
 - a. Using an interface, communicate with the Main Activity to pop the back stack, which should display the Genres fragment.
4. Clicking a row item should communicate with the Main Activity to:
 - a. Send the selected Book and replace the current fragment with the Book Details fragment.
 - b. Push the current fragment on the back stack.

Part 3 (10 Points): Book Details Fragment

This fragment is shown in Fig 1(c). The requirements are listed below:

1. This fragment should receive the Book object from the Books Fragment. The book information should be displayed as shown in Figure 1(c).
2. Clicking "Back" should:
 - a. Using an interface, communicate with the Main Activity to pop the back stack, which should display the Books fragment.